

Detailed Proposal for the Development and Deployment of an Autonomous Drone System on a Heavy Launch Vehicle for Habitable Planet Exploration and Orbital Repairs, Including Detailed Written Schematics and Multiple Deployment Options

Executive Summary

This proposal outlines an innovative space exploration system utilizing NASA's **Space Launch System (SLS)** as the primary launch vehicle, carrying **two or more autonomous drones**. The system focuses on the development of two types of drones designated **CAGS_F413**, where CAGS represents the initials of the creators (Camilo, Ángel, Gustavo, and Sara), and *F413* refers to Philippians 4:13 — *"I can do all things through Christ who strengthens me."*

The system includes a **primary repair and maintenance drone (CAGS_F413-R)**, equipped with robotic arms, multifunctional tools, an anchoring arm, solar panels, and ion thrusters for maneuvers in the vacuum of space.

The **secondary exploration drones (CAGS_F413-S)**, inspired by NASA's *Ingenuity* helicopter, are designed for initial deployment to map terrain, collect samples, detect signs of life, and scan for damage on the main rocket or nearby space stations. Any detected anomaly triggers the intervention of the repair drone.

As a scalability option, the proposal includes the **deployment of multiple SLS rockets (e.g., 3–5 units)**, each equipped with its own drone set. This would enable broader exploration coverage, greater operational redundancy, and parallel missions. The design is compatible with the SLS Block 1 configuration (total height: 98 meters; payload to LEO: up to 95 tons).

Estimated total costs for development, manufacturing, and the initial launch (single scenario) amount to approximately **\$3.2 billion USD**. For a 3-rocket deployment, the total would reach approximately **\$8.5 billion USD**.

The mission aligns with **NASA's Artemis objectives** and sustainable exploration goals, reducing human risk and extending the operational life of orbital assets.

Recommendations: Begin the prototyping phase at **NASA's Jet Propulsion Laboratory (JPL)**, with integration at the **Kennedy Space Center**. Assess multi-launch feasibility in coordination with the **SLS program for Artemis**.

1. Mission Description and Objectives

General Objectives

1. **Planetary Exploration:** Explore a habitable planet (e.g., Mars as baseline) to search for microbial life or water resources, collect samples for in-situ analysis, and enable potential sample return.
2. **Robotic Maintenance:** Monitor and repair damage on the primary launch vehicle and space stations (e.g., ISS modules or future lunar/martian habitats).
3. **Autonomy Demonstration:** Demonstrate autonomous operation in hostile space environments, minimizing human intervention and extending the life of critical assets.

Mission Scenario (Detailed)

Phase	Operation Description	Estimated Duration
Launch	The SLS Block 1 carries the drones inside an 8.4 m payload	T+0 to T+900

	fairing.	s
Initial Deployment	Upon reaching orbit or interplanetary trajectory, 2–3 exploration drones (EDs) are released first for reconnaissance.	T+1200 s
Operations	EDs perform reconnaissance flights, LIDAR/multispectral scans, and sample collection. They inspect the rocket and nearby stations for micrometeorite damage or thermal degradation.	2–5 years
DPRM-1 Intervention	Data is transmitted to DPRM-1. The unit is then released, maneuvers using ion thrusters, anchors to the damaged site, and executes repairs using specialized tools.	On Demand
Power	Continuous power via solar panels.	Continuous

2. Launch Vehicle and Payload Specifications

The **SDA-EPM** system is based on NASA's **SLS Block 1** due to its high deep-space payload capacity.

Characteristic	Specification	Notes
Total Height	98 m (322 ft)	
Core Stage Diameter	8.4 m (27.6 ft)	
Core Stage Dry Weight	85,275 kg (188,000 lb)	
Payload to LEO	95 tons (209,000 lb)	
Payload to TLI (Moon)	27–38 tons (59,500–83,700 lb)	Depends on upper stage configuration
Mission Configuration	SLS Block 1 with 8.4 m fairing	Accommodates drone deployment module
Total Drone Mass	~150 kg	<1% of LEO capacity

3. Autonomous Drone System Design

3.1. Main Repair and Maintenance Drone (CAGS_F413-R)

Designed for precise orbital maneuvers and complex repair operations.

Component	Technical Specification	Key Materials
Dimensions/Mass	2.5 m wingspan (extended); 100 kg	Al-Li 2219 alloy
Robotic Arms	2 units, 7 DOF each, 1.5 m	Titanium Grade 5 (Ti-6Al-4V)
Tools	Laser welder (50 W), Gripper (20 N), Drill (5 mm, 1000 RPM)	Steel/Ceramic
Anchoring Arm	2 m telescopic, electromagnetic hook (100 N)	Ti-6Al-4V
Propulsion	4 Ion Thrusters (NEXT type), 0.25 N each	Xenon propellant
Power	5 m ² deployable solar panels (>30% efficiency)	GaAs on Kapton
Processor	RAD750 radiation-hardened CPU	

3.2. Exploration and Scanning Drones (CAGS_F413-S)

Lightweight design optimized for atmospheric flight and data collection, scaled from *Ingenuity*.

Component	Technical Specification	Key Materials
Dimensions/Mass	1.2 m rotor span; 25 kg each	Carbon-reinforced polymer
Rotors	Counter-rotating, 4 blades per rotor, 2500 RPM	Epoxy composite
Flight Time	30 minutes per charge	Li-Ion batteries
Sensors	Multispectral cameras (Visible/IR/UV), LIDAR (100 m range)	Radiation-hardened ceramic
Sampling Arms	2 small arms (0.5 m), 5 N gripper, 500 g capacity	Al-Li alloy
Autonomous Range	1 km per flight	GPS/LIDAR navigation

4. Written Structural Schematics (Hierarchical)

The schematics follow **NASA-STD-5001 (Structural Design Standard)** to facilitate CAD model conversion.

Plan 1: SLS Block 1 – Drone Payload Configuration

- **Level 1:** Core Stage (LH₂ and LOX tanks, RS-25 engines)
- **Level 4:** Payload Fairing (8.4 m diameter)
 - Sublevel 4.1: Drone Deployment Module (Height: 4 m, Diameter: 2 m, Mass: 200 kg)
 - Sublevel 4.1.1: DPRM-1 Compartment
 - Sublevel 4.1.2: DE-1/DE-2/DE-3 Compartments

Plan 2: Repair Drone (DPRM-1)

- Central fuselage (1.2 m long, 0.4 m diameter, 40 kg)
- Robotic arms (1.5 m, 7 DOF)
- Anchoring arm (2 m)
- 4 Ion thrusters (0.25 N each)
- Operational sequence: Ion thrust → Anchoring → Tool activation (5–30 min)

Plan 3: Exploration Drones (DE-1/DE-2/DE-3)

- Central body (0.5 m height, 1.2 m rotor span, 25 kg)
- Counter-rotating rotors (2500 RPM)
- Sensors (LIDAR, HD/IR cameras)
- Sampling arms (500 g capacity)
- Flight cycle: Autonomous → Scan → Return → Recharge

5. Scalable Deployment Options

To maximize resilience and coverage, a **multi-launch deployment** (e.g., 3 SLS rockets) is proposed.

Factor	Single Launch (1 SLS)	Multiple Launch (3 SLS)
Exploration Coverage	Limited to one region	Broad, multi-biome coverage
Operational Redundancy	Low	High (mesh-linked missions)
Mission Duration	2–5 years	5+ years coordinated
Coordination	Local mission	Networked multi-mission
Artemis Integration	Standalone	Aligned with 2028–2030 multi-launch plans

6. Detailed Cost Breakdown (in Millions USD)

6.1. Single Scenario (1 SLS)

Category	Description	Estimated Cost	Notes
SLS Development	Engineering & initial testing	500	Boeing heritage contract
SLS Manufacturing	SLS Block 1 build	2,000	Includes RS-25 engines
Propellant	LH ₂ /LOX + boosters	50	Based on previous missions
Drone Development	DPRM + 2 DE prototypes	300	Similar to Curiosity + Ingenuity
Drone Manufacturing	Production of 3 units	150	Advanced materials
Propulsion & Power	Ion thrusters + solar panels	100	NEXT-based thrusters
Operations & Launch	Integration, testing, 2-year mission	100	DSN communications
TOTAL		3,200	—

6.2. Multi-Launch Scenario (3 SLS)

Category	Description	Estimated Cost	Notes
Development	Shared across missions	600	Economies of scale
SLS Manufacturing	3 Block 1 units	5,500	Bulk discounts
Propellant	3 launches	150	—
Drone Production	9 total units	400	Reduced unit cost
Propulsion & Power	3 drone sets	250	—
Operations	Multi-mission coordination	600	—
TOTAL		8,500	—

7. Operations, Materials, and Manufacturing Plan

Operations Plan

- **Transit Phase (Months 1–6):** Drones in hibernation; remote system checks.
- **Deployment (Month 6):** Sequential release of DE-1/DE-2 for surface/orbital exploration (50 missions per drone).
- **Monitoring & Repair:** Daily scans of the rocket/stations; alerts sent via laser link to DPRM-1.

Risks & Mitigations

- Propulsion failure → redundant ion thrusters
- Martian dust → antistatic coatings and sensor purge
- Communication loss → high-gain links and relay satellites

Materials & Fabrication

- **Main Materials:** Al-Li 2195 alloys, flexible GaAs solar cells, Titanium Grade 5
- **Shielding:** Kevlar + Nextel for micrometeorite protection
- **Manufacturing:**
 - SLS assembly: *Michoud Assembly Facility*
 - Drones: *JPL prototyping*, additive manufacturing (3D printing) for arms, integration at *Glenn Research Center*

8. Conclusions and Recommendations

This proposal presents a **realistic and scalable framework** for autonomous exploration and maintenance missions, aligned with NASA's vision of **sustainable exploration** and **reduced human risk** in deep space.

The multi-launch concept enhances mission impact without requiring major redesigns of the drone platform.

Recommendation: Immediate review by the **Artemis Program Team** for potential integration into **2028–2030 missions**, including detailed cost modeling for **3–5 sequential launches**.

References

- NASA SLS Fact Sheets and Reference Guides (2022–2023)
- NASA OIG Cost Assessment Reports (SLS Program)
- *Ingenuity* specifications and ion thruster concepts
- NASA Material Standards: *NASA-STD-6016* and *NASA-STD-5001*